Experiment 256: x-ray fluorescence

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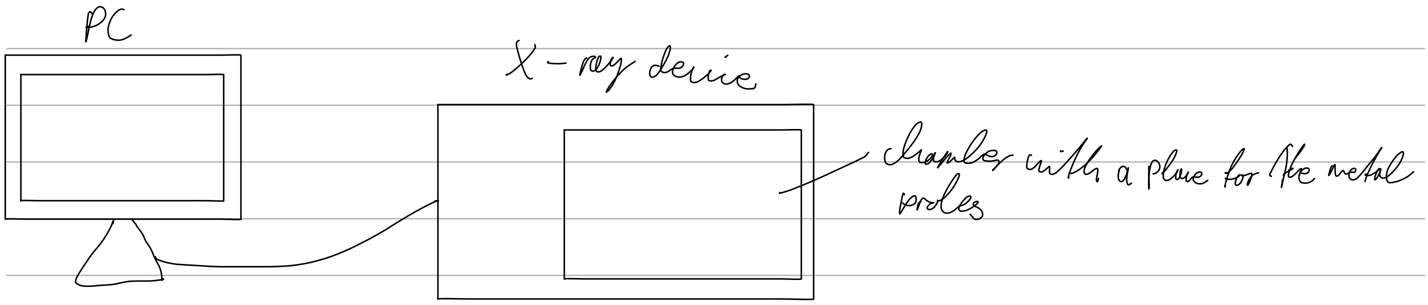
Equipment

- X-ray device with an X-ray detector
- Multichannel analyzer
- several samples of metal (Zr, Zn, Ni, Ti, Ag, Fe, Cu, Mo, 2 brass, stainless steel, magnet, unknown alloy)
- computer

Setup



Task 1: Calibration of the X-ray device

We set the measurement parameters as the following:

number of channels: 512
negative pulses
amplification factor: -2.5
measurement time: 180s
voltage: 35kV
current: 1mA

We also need to calibrate the x-axis. To do this, we fit a Gaussian curve to the $K\alpha$-Lines of Fe and Mo, of which we know the energies ($E_{\alpha,Fe} = 6.4$ keV, $E_{\alpha,Mo} = 17.48$ keV). We record their channels as $\mu_{\alpha,Fe} = 80$, $\sigma_{\alpha,Fe} = 2$, $\mu_{\alpha,Mo} = 216$ and $\sigma_{\alpha,Mo} = 2$, and can thus calibrate the x-axis.

Task 2: Measuring line energies as Gaussian curves

We make the following measurements:

Table 1: Energies of the Gaussian fits for every line

line	μ [keV]	σ [keV]
Zr, α	15.84	0,18
Zr, β	17,69	0,16
Zn, α	8,72	0,19
Zn, β	9,66	0,19
Fe, α	6,44	0,18
Fe, β	7,04	0,32
Mo, α	17,49	0,19
Mo, β	19,56	0,16
Ni, α	7,52	0,19
Ni, β	8,28	0,27
Cu, α	8,10	0,19
Cu, β	8,97	0,25
Ti, α	4,61	0,17
Ti, β	–	–
Ag, α	21,84	0,19
Ag, β	24,49	0,23

There was barely a bump in the curve, and we could not fit a satisfactory Gaussian

Measurement order: Cu, Ag, Zn, Ti, Zr, Ni, Mo, Fe

Task 3: Compositional analysis of alloys

We measure the spectrums of our alloy samples, although we do not analyse them numerically. After doing so, we use the programs built-in values for an elements X-ray lines to analyse their composition:

stainless steel: Fe, Ti, V, Mn
brass: Cu, Zn
magnet: Y, Cu, Zn, Ir, Mn
unknown alloy: Cu, Ni, Hg (?)